

This page is the record for studying linear algebra, Hoffman & Kunze.

Assume $T: V \rightarrow V$, V is finite dimensional vector space.

Lemma. Let $\alpha \in V$ be an eigenvector, $T\alpha = c\alpha$.
For any $f \in F[x]$, $f(T)\alpha = f(c)\alpha$.

Thm. Let $c_1, \dots, c_k \in F$ be distinct eigenvalues. Then,

$$\dim \sum_{i=1}^k E_{c_i} = \sum_{i=1}^k \dim E_{c_i}$$

Moreover, if β_i ($1 \leq i \leq k$) is basis for E_{c_i} , then $\bigcup_{i=1}^k \beta_i$ is basis for $\sum_{i=1}^k E_{c_i}$
Proof. Since $W_1 + W_2 \leq V$ if $W_1, W_2 \leq V$ and $\langle \beta_1 \rangle + \langle \beta_2 \rangle = \langle \beta_1 \cup \beta_2 \rangle$

$$\sum_{i=1}^k E_{c_i} = \sum_{i=1}^k \text{span}(\beta_i) = \text{span}\left(\bigcup_{i=1}^k \beta_i\right)$$

Meanwhile, for each $1 \leq i \leq k$, let $B_i \in E_{c_i}$. Suppose that

$$B_1 + B_2 + \dots + B_k = 0.$$

For any polynomial $f \in F[x]$,

$$0 = f(T)(0) = f(T)(B_1 + \dots + B_k) = f(T)(B_1) + \dots + f(T)(B_k) = f(c_1)B_1 + \dots + f(c_k)B_k$$

Using Lagrange's interpolation, choose f_i ($1 \leq i \leq k$) s.t.

$$f_i(c_j) = \delta_{ij}.$$

Now, $0 = \sum_{i=1}^k f_i(c_j) B_i = B_j$, for each $1 \leq j \leq k$. In summary,

$$B_1 + \dots + B_k = 0 \quad (B_i \in E_{c_i}) \Rightarrow B_i = 0 \quad (1 \leq i \leq k)$$

Now, the set $\bigcup_{i=1}^k \beta_i$ is linearly independent, because

if $\sum_{i=1}^k \sum_{j=1}^{r_i} a_{i,j} \cdot B_{i,j} = 0$ ($B_{i,j} \in \beta_i$), then $\sum_{j=1}^{r_i} a_{i,j} B_{i,j} = 0$ ($1 \leq i \leq k$),

thus all $a_{i,j} = 0$. Now Proved.

Thm. Let $T: V \rightarrow V$ be a linear operator on a finite-dimensional space V .

Let $c_1, \dots, c_k$ be the distinct eigenvalues of T . TFAE:

a). T is diagonalizable.

b). $\text{ch}(T) = \prod_{i=1}^k (x - c_i)^{d_i}$ where $d_i = \dim E_{c_i}$.

c). $\dim V = \sum_{i=1}^k \dim E_{c_i}$

Minimal Polynomial.

Thm. The characteristic and minimal polynomials for T have the same roots.

Proof. Let p be the minimal polynomial for T .

Suppose that $c \in F$ is root of p . That is, $p(c) = 0$. Then,

$$p = (x - c) \cdot q \quad (\deg q < \deg p)$$

for some $q \in F[x]$. Since p is the minimal, $q(T) \neq 0$. Take $\beta \in V$ s.t. $q(T)\beta \neq 0$. Then, $0 = p(T)\beta = (T - cI) \cdot q(T)\beta$. Thus $T \cdot q(T)\beta = c\beta$, c is eigenvalue.

Conversely, let $c \in F$ be an eigenvalue of T . That is, $T\beta = c\beta$ for some nonzero $\beta \in V$.

Now, $0 = p(T)\beta = p(c)\beta$ and since β is nonzero, $p(c) = 0$.

Note; the Minimal polynomial is NOT necessarily irreducible; actually,

$$\mathbb{Q}_T: F[x] \rightarrow \mathcal{L}(V, V); f(x) \mapsto f(T)$$

the minimal polynomial is the generator of the ideal $\ker \mathbb{Q}_T$, where $\mathbb{Q}_T$ is the ring-homomorphism. Since $\mathcal{L}(V, V)$ is a ring under $(+, \cdot)$ and it contains a zero divisor; for instance, $\dim V = 2$,

$$\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \cdot \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} = 0$$

Thus, even if $p \in F[x]$ is the minimal polynomial,

$$0 = p(T) = f(T) \cdot g(T) \quad \text{for some } \deg f, \deg g < \deg p$$

can be happen.

Invariant.

(199.P)

Let $T, U: V \rightarrow V$ be linear operators, and commute.

Both $\text{Im } U$ and $\text{Ker } U$ are invariant under T , because:

$$y \in \text{Im } U \Rightarrow y = U(x) \text{ for some } x \in V \Rightarrow T(y) = T(U(x)) = U(T(x)) \in \text{Im } U$$

$$y \in \text{Ker } U \Rightarrow U(y) = 0 \Rightarrow 0 = T(U(y)) = U(T(y)) \Rightarrow T(y) \in \text{Ker } U.$$

Since T commutes with $g(T)$, where $g \in F[x]$,

$\text{Ker } g(T)$ and $\text{Im } g(T)$ are invariant under T .

Lemma. Let $W \subseteq V$ be an invariant subspace for $T: V \rightarrow V$. Then,

$\text{ch}(T_W)$ divides $\text{ch}(T)$. Moreover, the minimal polynomial of T_W divides minimal polynomial of T .

Proof. By the calculation,

$$[T]_{\beta} = \begin{bmatrix} [T_W]_{\beta'} & B \\ 0 & C \end{bmatrix} \quad \text{where } \beta \text{ and } \beta' \text{ are basis for } V \text{ and } W, \text{ respectively,} \\ \text{and } B' \subset \beta$$

$$\text{Now, } \text{ch}(T) = \det(xI - [T]_{\beta}) = \det(xI - [T_W]_{\beta'}) \cdot \det(xI - C) \\ = \text{ch}(T_W) \cdot \det(xI - C),$$

thus the first assertion is proved.

Conductor.

Def. Let W be an invariant subspace for $T: V \rightarrow V$, and $\alpha \in V$ be a vector. Define T -Conductor of α into W as;

$$S_T(\alpha; W) \stackrel{\text{def}}{=} \{g \in F[x] \mid g(T)(\alpha) \in W\}.$$

In particular, if $W = \{0\}$, the Conductor is called the T -annihilation of α .

Lemma. The Conductor $S(\alpha; W)$ is an Ideal in $F[x]$.

Proof. First, if $T[W] \subseteq W$, then $g(T)[W] \subseteq W$, because for any $w \in W$,
 $\beta \in W \Rightarrow T^n(\beta) = T^{n-1}(T(\beta)) = \dots = T(T^{n-1}(\beta)) \in W$.

And since $W \subseteq V$, it is closed under the addition and scalar multiplication, $g(T)[W] \subseteq W$. Now, to show $S(\alpha; W)$ is an Ideal, first show that it is a subspace. For any $f, g \in S(\alpha; W)$ and $c \in F$, $(cf + g)(T) = cf(T) + g(T)$, therefore

$$(cf + g)(T)(\alpha) = cf(T)(\alpha) + g(T)(\alpha) \in W$$

Thus it is a subspace. Moreover, for any $f \in F[x]$ and any $g \in S(\alpha; W)$,

$$(fg)(T)(\alpha) = f(T) \underbrace{g(T)(\alpha)}_{\in W} \in W.$$

Thus $fg \in S(\alpha; W)$. $\square$

Note: Given the linear operator T and T -invariant subspace W , and any vector $\alpha \in V$,

$$Q: F[x] \rightarrow V/W : f(x) \mapsto f(T)(\alpha) + W$$

is $F[x]$ -module homomorphism, so $S_T(\alpha; W) = \ker Q$.

Lemma. If the minimal polynomial $P(x) \in F[x]$ for T factors as:

$$P = (x - c_1)^{r_1} \dots (x - c_k)^{r_k}, \quad c_i \in F.$$

Condition

and let W be a proper subspace of V which is invariant under T , then there is a vector $\alpha \in V$ s.t.

$$\alpha \notin W$$

results

2) For some eigenvalue $c \in F$ of T , $(T - cI)(\alpha) \in W$.

Proof. Let $\beta \in V \setminus W$ be given. Let $q \in F[x]$ be the T -annihilator of β into W .

That is, $q(T)(\beta) \in W$. Since $(q(x) = S_T(\alpha; W))$ and $P(T)(\alpha) = 0 \in W$, $P \in S_T(\alpha; W)$ therefore q divides P . And, since $\beta \notin W$, q must be not constant. Now,

$$q = (x - c_1)^{e_1} \dots (x - c_k)^{e_k}, \quad 0 \leq e_i \leq r_i, \text{ at least one } e_i \text{ is positive.}$$

Let j be a integer s.t. e_j is positive. Then,

$$q = (x - c_j)^{e_j} \cdot h \quad \text{for some } h \in F[x].$$

Since q is the generator of $S_T(\alpha; W)$, $h(T)\beta \notin W$. But,

$$(T - c_j I)(h(T)\beta) = q(T)(\beta) \in W.$$

Thm. Let V be a finite dimensional vector space over F , and $T: V \rightarrow V$ be a linear.

T is triangulable if and only if minimal polynomial of T splits over F .

Proof. If T can be represented by triangular, then clear, since:

the determinant of triangle is product of diagonals. Conversely, suppose that

$$P(x) = (x - c_1)^{r_1} \dots (x - c_k)^{r_k}$$

is the minimal polynomial of T . Since P shares the roots with $\chi(T)$, c_i 's are eigenvalues.

Choose $\alpha_1 \in V$ s.t. $\alpha_1 \notin \{0\}$ and $(T - c_1 I)(\alpha_1) \in \{0\}$, for some eigenvalue $c \in F$.

Choose $\alpha_2 \in V$ s.t. $\alpha_2 \notin \langle \alpha_1 \rangle$ and $(T - c_2 I)(\alpha_2) \in \langle \alpha_1 \rangle$, //

$$T(\alpha_2) = c_1 \alpha_1 + c_2 \alpha_2$$

Thm. Let V be a vector space over F .

For any non-zero linear functional $f \in V^*$, $\ker f$ is a hyperplane of V .

Conversely, every hyperplane in V is the kernel for some linear functional on V .

Proof. Let $f \in V^*$ be a non-zero linear functional. Since f is non-zero, choose $\alpha \in V \setminus \ker f$ (or $f(\alpha) \neq 0$). We shall show that: $V = \ker f + \langle \alpha \rangle$.

Let $\beta \in V$ be given. Put $c = f(\beta)/f(\alpha)$. (It can be defined because $f(\alpha) \neq 0$).

Then, $f(\beta - c\alpha) = f(\beta) - cf(\alpha) = 0$, thus $\beta - c\alpha \in \ker f$, or $\beta \in \ker f + \langle \alpha \rangle$.

Conversely, let $N \subseteq V$ be the hyperplane. Let $\alpha \in V \setminus N$.

Since N is the hyperplane, $N + \langle \alpha \rangle = V$. Therefore, for each $\beta \in V$,

$$\beta = \gamma + c\alpha \quad \text{for some } \gamma \in N, c \in F.$$

This γ and c are uniquely determined, because $N \cap \langle \alpha \rangle = \{0\}$.

Define $g: V \rightarrow F$ as $g(\beta) = c$. Then, $\ker g = N$.

$$\begin{pmatrix} a_1 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & a_3 \end{pmatrix}^2 = \begin{pmatrix} a_1^2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & a_3^2 \end{pmatrix}$$

Lemma. Let V be a fin. dim v.s over F , and let $T: V \rightarrow V$ be a linear operator. Suppose that the minimal polynomial of T splits over F as:

$$p = (x - c_1)^{r_1} \cdots (x - c_k)^{r_k} \quad (c_i \in F)$$

Then, for any proper T -invariant subspace $W < V$, there exists a vector $\alpha \in V$ s.t. $\alpha \notin W$ and for some c_i , $(T - c_i I)(\alpha) \in W$.

Proof. Let $\beta \in V \setminus W$ be given, and let $q \in F[x]$ be a T -conductor of β into W . Then, q divides p . And, since $\beta \notin W$, q is not constant. Therefore

$$q = (x - c_1)^{e_1} \cdots (x - c_k)^{e_k} \quad (0 \leq e_i \leq r_i)$$

and at least one of e_i is positive. Pick i s.t. $e_i > 0$. Then,

$$q = (x - c_i) \cdot h \quad \text{where } h = q / (x - c_i) \in F[x].$$

Since q is T -conductor of β , $h(T)(\beta) \notin W$, but $(T - c_i I)(h(T)(\beta)) = q(T)(\beta) \in W$.

Thm. Let V be a finite dimensional vector space over F , and $T: V \rightarrow V$ be a linear. T is Triangularizable if and only if minimal polynomial of T splits over F .

Proof. If T is triangularizable with respects to a basis β , then the characteristic polynomial is

$$f(x) = \det([T]_{\beta} - xI) = \prod_{i=1}^n (A_{ii} - x) \quad \text{where } A_{ii} \text{ is } i\text{th diagonal element of } [T]_{\beta}$$

thus, proved. Conversely, choose $\alpha_1 \in V$ s.t. $\alpha_1 \notin \{0\}$, $(T - c_1 I)(\alpha_1) \in \{0\}$ for some eigenvalue $c_1 \in F$ of T . Choose $\alpha_2 \in V$ s.t. $\alpha_2 \notin \langle \alpha_1 \rangle$, $(T - c_2 I)(\alpha_2) \in \langle \alpha_1 \rangle$, it is possible because $\langle \alpha_1 \rangle$ is T -invariant. Again, choose $\alpha_3 \in V$ s.t. $\alpha_3 \notin \langle \alpha_1, \alpha_2 \rangle$, $(T - c_3 I)(\alpha_3) \in \langle \alpha_1, \alpha_2 \rangle$. It is possible again, $T(\alpha_2) \in \langle \alpha_1, \alpha_2 \rangle$.

Now, we obtain $\{\alpha_1, \dots, \alpha_n\}$ s.t.

$$T(\alpha_1) = c_1 \alpha_1$$

$$T(\alpha_2) = a_{21} \alpha_1 + c_2 \alpha_2$$

$\vdots$

$$T(\alpha_n) = a_{n1} \alpha_1 + a_{n2} \alpha_2 + \dots + c_n \alpha_n$$

Thm: Let V be a fin. d'im. v.s over F , and $T: V \rightarrow V$ be a linear operator.
 T is diagonalizable if and only if minimal polynomial of T splits and separable.

Proof. Suppose that T is diagonalizable. Let $c_1, \dots, c_k$ be distinct eigenvalues.
 Then, for any eigenvector $d \in V$, there is a $c_i \in F$ s.t. $(T - c_i I)(d) = 0$.
 Therefore, for $p = (x - c_1) \dots (x - c_k)$,

$$p(T) = (T - c_1 I) \dots (T - c_k I)(d) = 0$$

because $\{T - c_1 I, \dots, T - c_k I\}$ commute. Since T is diagonalizable,
 for any vector of V can be represented by linear combination of eigenvectors,
 $p(T) = 0$

Thus proved. Conversely, let W be a spanned subspace by all of eigenvectors of T .
 Suppose that $W \neq V$. By the construction, W is T -invariant. Therefore, by the Lemma,
 there is $\alpha \in V \setminus W$ and eigenvalue $c_0 \in F$ s.t. $\beta = (T - c_0 I)(\alpha) \in W$.
 Since $\beta \in W$, $\beta = \beta_1 + \dots + \beta_k$ where $T(\beta_i) = c_i \beta_i$, $1 \leq i \leq k$. Therefore, for any $h \in F[x]$

$$h(T)(\beta) = \sum_{i=1}^k h(c_i) \cdot \beta_i \in W.$$

Put $p = (x - c_0)g$. Also, $g - g(c_0) = (x - c_0)h$. Then,

$$g(T)(\alpha) - g(c_0)(\alpha) = h(T)(T - c_0 I)(\alpha) = h(T)(\beta) \in W$$

And, $0 = p(T)(\alpha) = (T - c_0 I)g(T)(\alpha) = g(T)(T - c_0 I)(\alpha) \in W$.

Therefore $g(c_0)(\alpha) \in W$, and since $\alpha \notin W$, $g(c_0) = 0$. Contradicts to $p = (x - c_0)^2 g$.

Lemma. Let $f, g: V \rightarrow F$ be linear functionals on V .

$g = c \cdot f$ for some $c \in F$ if and only if $\ker f \leq \ker g$.

Proof. If $f = 0$, then trivial. Assume $f \neq 0$. Then, $\ker f$ is the hyper-space of V .
Let $\alpha \in V$ s.t. $f(\alpha) \neq 0$. put $c = g(\alpha)/f(\alpha)$. Then, $g - cf$ is a linear functional s.t.
 $\ker f \leq \ker(g - cf)$ and $\alpha \in \ker(g - cf)$. Thus $\ker(g - cf) = V$, or $g - cf = 0$.

Thm. Let $g, f_1, \dots, f_r$ be linear functionals on V . Then,

g is linear combination of $\{f_1, \dots, f_r\}$ iff $\bigcap_{i=1}^r \ker f_i \leq \ker g$.

Proof. If $g = \sum_{i=1}^r c_i f_i$ for some $c_i \in F$, then for any $\alpha \in \bigcap_{i=1}^r \ker f_i$, $g(\alpha) = \sum_{i=1}^r c_i f_i(\alpha) = 0$.
Conversely, we shall use the Mathematical induction for $r \in \mathbb{N}$.

By the previous Lemma, if $r=1$, it is proved. Suppose that it is held for $r=k-1$.

Let $f_1, \dots, f_k$ be linear functionals on V s.t.

$$\bigcap_{i=1}^k \ker f_i \leq \ker g.$$

Let $g, f_1, \dots, f_{k-1}$ be restricted by $\ker f_k$. That is,

$$f_i|_{\ker f_k}: \ker f_k \rightarrow F; \alpha \mapsto f_i(\alpha). \quad (i=0, \dots, k-1, f_0 = g)$$

And, let $\alpha \in \ker f_k$ s.t. for all $1 \leq i \leq k-1$, $f_i(\alpha) = 0$.

That is, $\alpha \in \bigcap_{i=1}^k \ker f_i$ and therefore $g|_{\ker f_k}(\alpha) = 0$, by the hypothesis.

By the assumption of the induction,

$$g|_{\ker f_k} = \sum_{i=1}^{k-1} c_i f_i|_{\ker f_k} \quad \text{for some } c_i \in F$$

Put $h = g - \sum_{i=1}^{k-1} c_i f_i$. Now, for any $\alpha \in \ker f_k$, $h(\alpha) = 0$, or $\ker f_k \leq \ker h$.

The previous Lemma gives that $h = c_k f_k$ for some $c_k \in F$. Now, $g = \sum_{i=1}^k c_i f_i$.

Proof.

$$\begin{array}{ccccc}
 V & \xrightarrow{\alpha_i \mapsto f_i} & V^* & \longrightarrow & V^{**} \\
 T \downarrow & & \uparrow T^t & & \downarrow T^{tt} \\
 W & \xrightarrow{\beta_i \mapsto g_i} & W^* & \longrightarrow & W^{**}
 \end{array}$$

$$\dim \ker f = \dim V - 1 \quad \text{where } f \in V^*.$$

$$\ker f := \{\alpha \in V \mid f(\alpha) = 0\}$$

$$S^0 := \{f \in V^* \mid f(\alpha) = 0 \quad \forall \alpha \in S\}.$$

$$\bigcap_{i=1}^r \ker g_i \subseteq W$$

$$\begin{aligned}
 W &= \ker f_{k+1} \cap \ker f_{k+2} \cap \dots \cap \ker f_n. \\
 &= F^{n-k}
 \end{aligned}$$

$$\begin{array}{c}
 A_{11}x_1 + \dots + A_{1n}x_n = 0 \\
 \vdots \\
 A_{m1}x_1 + \dots + A_{mn}x_n = 0
 \end{array}$$

$$W \subseteq \ker f \Leftrightarrow \bigcap \ker g_i \subseteq \ker f.$$

For each $1 \leq i \leq m$, define

$$f_i: F^n \rightarrow F; (x_i) \mapsto \sum_{j=1}^n A_{ij}x_j$$

$$\ker f_1 \cap \ker f_2 \cap \dots \cap \ker f_m$$

$W \subseteq V$, $\{g_1, \dots, g_r\}$ is basis for W^0 .

$$W = \bigcap_{i=1}^r \ker g_i$$

Proof. $W \subseteq \bigcap \ker g_i$ is trivial, because $W \subseteq \ker g_i$ for all $g_i \in W^0$.

Conversely, let $\{f_1, \dots, f_m\} \subseteq V^*$ s.t. $W = \ker f_1 \cap \dots \cap \ker f_m$. Then, for each $1 \leq i \leq m$,

$$\begin{aligned}
 W \subseteq \ker f_i &\Leftrightarrow f_i \in W^0 \Leftrightarrow f_i \text{ is linear combination of } \{g_1, \dots, g_r\} \\
 &\Leftrightarrow \bigcap_{i=1}^r \ker g_i \subseteq \ker f_i
 \end{aligned}$$

This proves.

Consider a system of linear equation;

$$\begin{array}{ccccccc} A_{11}x_1 & + & \dots & + & A_{1n}x_n & = & 0 \\ \vdots & & & & \vdots & & \vdots \\ A_{m1}x_1 & + & \dots & + & A_{mn}x_n & = & 0 \end{array}$$

Define $f_i: F^n \rightarrow F: (x_j) \mapsto (A_{ij} \cdot x_j)$ for each $1 \leq i \leq m$.

We want to find a subspace: $\ker f_1 \cap \dots \cap \ker f_m$. (has dimension $\leq n-m$)

1). If $m < n$, it has a non-trivial solution.

2). If $m = n$ and

Def. Let $W_i \leq V$.

If $\alpha_1 + \dots + \alpha_k = 0$ ($\alpha_i \in W_i$) $\Rightarrow \alpha_i = 0$,

then W_i 's are independent.

Lemma. Let $W = \sum W_i$ TFAE

1) W_i 's are independent.

2) For each $2 \leq j \leq k$, $W_j \cap (W_1 + \dots + W_{j-1}) = \{0\}$.

3) If β_i is basis for W_i , then $\beta_1 \cup \dots \cup \beta_k$ is basis for W .

proof. 1) $\Rightarrow$ 2)

Let $\alpha \in W_j \cap \sum_{i=1}^{j-1} W_i$. Then, $\alpha = \alpha_j = \alpha_1 + \dots + \alpha_{j-1}$. Then,

$$\alpha_1 + \dots + \alpha_{j-1} - \alpha_j = 0 \Rightarrow \alpha_j = 0.$$

2) $\Rightarrow$ 1). Suppose $\alpha_1 + \dots + \alpha_k = 0$ ($\alpha_i \in W_i$). If not all zero, let j be the largest integer st $\alpha_j \neq 0$. Then, $0 = \alpha_1 + \dots + \alpha_j \Rightarrow \alpha_j = -\alpha_1 - \dots - \alpha_{j-1} \in W_j \cap \left\{ \sum_{i=1}^{j-1} W_i \right\}$.

Def. A linear operator $E: V \rightarrow V$ is called a projection if $E^2 = E$.

• $x \in \text{Im } E \iff E(x) = x$

Proof. $x \in \text{Im } E \Rightarrow x = E(\alpha)$ for some $\alpha \in V \Rightarrow E(x) = E^2(\alpha) = E(\alpha) = x$. Converse is trivial.

• $V = \text{Im } E \oplus \text{Ker } E$

Proof. Given $x \in V$, $x = E(x) + (x - E(x))$. Clearly $E(x) \in \text{Im } E$, and

$$E(x - E(x)) = E(x) - E^2(x) = 0 \Rightarrow x - E(x) \in \text{Ker } E$$

And, let $y \in \text{Im } E \cap \text{Ker } E$ be given. Then, $E(y) = y$ and $E(y) = 0$, thus $y = 0$.

Thm. Let $V = W_1 \oplus W_2$. Then, there is one and only one projection $E: V \rightarrow V$ s.t.

$$\text{Im } E = W_1 \quad \text{and} \quad \text{Ker } E = W_2$$

Proof. Ask a teacher on the street.

Let E be a projection. For β_1 , basis for $\text{Im } E$, and β_2 , basis for $\text{Ker } E$,

$$[E]_{\beta_1 \cup \beta_2} = \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix} \quad \text{where } r = \text{rank } E.$$

Thm. Let V be a fin. dim. v.s over F .

If $V = \bigoplus_{i=1}^k W_i$, then there exist k linear operators E_i on V s.t.

$$E_i \circ E_j = \delta_{ij} \cdot E_i, \quad I = \sum_{i=1}^k E_i, \quad W_i = \text{Im } E_i \quad (1 \leq i \leq k). \quad \dots (*)$$

Conversely, if $E_1, \dots, E_k$ are k linear operators on V which satisfy $(*)$, then

$$V = \bigoplus_{i=1}^k W_i$$

Proof. Suppose $V = \bigoplus_{i=1}^k W_i$. For each $1 \leq i \leq k$, define $E_i: V \rightarrow V: \sum_{j=1}^k \alpha_j \mapsto \alpha_i$. proved.

Conversely, suppose that $E_1, \dots, E_k$ are linear operators on V satisfying $(*)$.

Given $\alpha \in V$, $\alpha = E_1(\alpha) + \dots + E_k(\alpha) \in W_1 + \dots + W_k$. And, to show uniqueness such expression,

let $\alpha = \alpha_1 + \dots + \alpha_n$, where $\alpha_i \in W_i$. And, let $\alpha_i = E_i(\beta_i)$, for each $1 \leq i \leq k$.

Now, for each i , $1 \leq i \leq k$,

$$E_j(\alpha) = \sum_{i=1}^k E_j(\alpha_i) = \sum_{i=1}^k (E_j \circ E_i)(\beta_i) = E_j^2(\beta_j) = E_j(\beta_j) = \alpha_j$$

Thus Proved.

Thm. Let V be a fin. dim. Vs and $T: V \rightarrow V$ be a linear operator.

Let $E_1, \dots, E_k$ be projection on V s.t. $E_i \circ E_j = \delta_{ij} \cdot E_i$, $I = \sum_{i=1}^k E_i$.

Then, each $\text{Im } E_i$ be invariant under T if and only if $T \circ E_i = E_i \circ T$, for all $1 \leq i \leq k$.

Proof. Suppose that T commutes with each E_i . Then, for any $\alpha \in \text{Im } E_i$,

$$T(\alpha) = T(E_i(\alpha)) = E_i(T(\alpha)) \in \text{Im } E_i.$$

Thus proved.

Conversely, suppose that each $\text{Im } E_i$ is invariant under T . Given $\alpha \in V$, by the assumption,

$$\alpha = E_1(\alpha) + \dots + E_k(\alpha) \Rightarrow T(\alpha) = T(E_1(\alpha)) + \dots + T(E_k(\alpha))$$

And since each $\text{Im } E_i$ is T -invariant, $T(E_i(\alpha)) = E_i(\beta_i)$ for some $\beta_i \in V$. Now,

$$(E_j \circ T \circ E_i)(\alpha) = (E_j \circ E_i)(\beta_i) = \delta_{ij} \cdot E_i(\beta_i)$$

$$\text{Finally, } (E_j \circ T)(\alpha) = (E_j \circ T \circ E_1)(\alpha) + \dots + (E_j \circ T \circ E_k)(\alpha) = E_j(\beta_j) = (T \circ E_j)(\alpha). \quad \square$$

Thm. Let T be a linear operator on fin. dim V .

If T is diagonalizable with distinct eigenvalues $C_1, \dots, C_k$, then there are linear operators $E_1, \dots, E_k$

$$T = C_1 E_1 + \dots + C_k E_k, \quad I = E_1 + \dots + E_k, \quad E_i \circ E_j = \delta_{ij} \cdot E_i, \quad \text{Im } E_i = \ker(T - C_i I).$$

Conversely, if there are linear operators E_i and distinct scalars C_i satisfying the condition above, then T is diagonalizable and has C_i 's as eigenvalues.

Proof. Suppose T is diagonalizable with distinct eigenvalues $C_1, \dots, C_k \in F$. Then,

$$V = \ker(T - C_1 I) \oplus \dots \oplus \ker(T - C_k I)$$

Define E_i be a projection onto $\ker(T - C_i I)$. Then, given $\alpha \in V$,

$$\alpha = E_1(\alpha) + \dots + E_k(\alpha) \quad \text{and so}$$

$$T(\alpha) = T(E_1(\alpha)) + \dots + T(E_k(\alpha)) = C_1 E_1(\alpha) + \dots + C_k E_k(\alpha) = (C_1 E_1 + \dots + C_k E_k)(\alpha).$$

Conversely,

[Thm. Primary decomposition Theorem.

Let T be a linear operator on fin. dim v.s V over a field F .

Let P be the minimal polynomial for T , with factorization

$$P = p_1^{r_1} \cdots p_k^{r_k} \quad (p_i \text{ are irreducibles, } r_i \in \mathbb{N}).$$

Then, for $W_i = \ker p_i(T)^{r_i}$ for each $(1 \leq i \leq k)$,

1). $V = W_1 \oplus \cdots \oplus W_k$

2). Each W_i is T -invariant.

3). $T|_{W_i}$ has minimal polynomial $p_i^{r_i}$.

Proof. For each $1 \leq i \leq k$, define $f_i = \prod_{j \neq i} p_j^{r_j}$. Since $p_1, \dots, p_k$ are distinct prime elements, so $f_1, \dots, f_k$ are relatively prime. That is,

$$(f_1, \dots, f_k) = (f_1) + \cdots + (f_k) = 1. \quad \text{+ More specifically, } (f_1, f_2, \dots, f_m) = \left(\prod_{j=1}^m p_j^{r_j} \right)$$

Now, there are polynomials $g_1, \dots, g_k$ s.t.

$$\sum_{i=1}^k f_i g_i = 1.$$

Moreover, $f_i f_j$ is divided by p if $i \neq j$. Let $E_i = (f_i g_i)(T)$, for each i . Then,

$$E_1 + \cdots + E_k = I \quad \text{and} \quad E_i \circ E_j = 0 \quad \text{if } i \neq j,$$

the second assertion is proved by

$$E_i \circ E_j = (f_i g_i)(T) \circ (f_j g_j)(T) = (g_i g_j)(T) \circ (f_i f_j)(T) = 0 \quad \text{if } i \neq j.$$

Therefore, these $E_1, \dots, E_k$ are projections. Moreover $\text{Im } E_i = W_i = \ker p_i(T)^{r_i}$, because let $\alpha \in \text{Im } E_i$. Then, $\alpha = E_i(\alpha)$, and so

$$p_i(T)^{r_i}(\alpha) = p_i(T)^{r_i} \circ E_i(\alpha) = p_i(T)^{r_i} \circ f_i(T) \circ g_i(T)(\alpha) = 0.$$

Conversely, let $\alpha \in \ker p_i(T)^{r_i}$. If $j \neq i$, then $f_j g_j$ is divided by $p_i^{r_i}$, and so

$$f_j(T) \circ g_j(T)(\alpha) = 0.$$

And thus, $E_i(\alpha) = \alpha$, $\alpha \in \text{Im } E_i$.

Inner product. Let $F = \mathbb{R}$ or $F = \mathbb{C}$.

Def. Inner product.

Let V be a vector space over F . A map

$$\langle \cdot, \cdot \rangle: V \times V \rightarrow F; (\alpha, \beta) \mapsto \langle \alpha, \beta \rangle$$

is called 'inner product' if it satisfies the following: for any $\alpha, \beta, \gamma \in V$ and $c \in F$,

1) $\langle c\alpha + \beta, \gamma \rangle = c\langle \alpha, \gamma \rangle + \langle \beta, \gamma \rangle$

2) $\langle \alpha, \beta \rangle = \overline{\langle \beta, \alpha \rangle}$

3) If $\alpha \neq 0$, then $\langle \alpha, \alpha \rangle > 0$

Lemma. An orthogonal set of non-zero vectors is linearly independent.

Proof. Let $S \subseteq V \setminus \{0\}$ be an orthogonal set.

Suppose that $\alpha_1, \dots, \alpha_n \in S$ are distinct vectors s.t.

$$\beta = c_1 \alpha_1 + \dots + c_n \alpha_n \text{ for some } c_i \in F.$$

Then, for each $1 \leq k \leq n$,

$$\langle \beta, \alpha_k \rangle = \langle \sum c_i \alpha_i, \alpha_k \rangle = \sum c_i \langle \alpha_i, \alpha_k \rangle = c_k \langle \alpha_k, \alpha_k \rangle \Rightarrow c_k = \frac{\langle \beta, \alpha_k \rangle}{\langle \alpha_k, \alpha_k \rangle}$$

Since each $\alpha_k \neq 0$, if $\beta = 0$, then $c_1 = \dots = c_n = 0$.

Furthermore, if $\beta \in V$ is linear combination of $\alpha_1, \dots, \alpha_n \in S$,

$$\beta = \sum_{i=1}^n \frac{\langle \beta, \alpha_i \rangle}{\|\alpha_i\|^2} \cdot \alpha_i.$$

Gram-Schmidt process.

Let V be a fin. dim. inn. prod. space, let $\{\beta_1, \dots, \beta_n\} \subseteq V$ be a linearly independent subset.

Then, there exists an orthogonal subset $\{\alpha_1, \dots, \alpha_n\} \subseteq V$ s.t.

$$\text{Span}\{\alpha_1, \dots, \alpha_n\} = \text{Span}\{\beta_1, \dots, \beta_n\}.$$

Proof. Using mathematical induction for n . For $n=1$, take $\alpha_1 = \beta_1$, then clear.

Now, suppose that for $m \in \mathbb{N}$, the statement holds. Let $\{\beta_1, \dots, \beta_{m+1}\}$ be lin. independent, and

$$\alpha_{m+1} = \beta_{m+1} - \sum_{i=1}^m \frac{\langle \beta_{m+1}, \alpha_i \rangle}{\|\alpha_i\|^2} \alpha_i \text{ where } \alpha_1, \dots, \alpha_m \text{ are obtained by hypothesis.}$$

Then $\alpha_{m+1} \neq 0$, because β_{m+1} is independent with $\{\alpha_1, \dots, \alpha_m\}$. Moreover, for $1 \leq j \leq m$,

$$\langle \alpha_{m+1}, \alpha_j \rangle = \langle \beta_{m+1}, \alpha_j \rangle - \sum_{i=1}^m \frac{\langle \beta_{m+1}, \alpha_i \rangle}{\|\alpha_i\|^2} \cdot \langle \alpha_i, \alpha_j \rangle = \langle \beta_{m+1}, \alpha_j \rangle - \langle \beta_{m+1}, \alpha_j \rangle = 0.$$

Now, $\{\alpha_1, \dots, \alpha_{m+1}\}$ is orthogonal, non-zero set, thus linearly independent by the Lemma.

Moreover, it is subset of $\text{Span}\{\beta_1, \dots, \beta_{m+1}\}$, the replacement Theorem gives result.

Orthogonal Projection.

Thm. Let W be a subspace of an inner prod. space V , and let $\beta \in V$.

- 1). A vector $\alpha \in W$ satisfies $\forall r \in W, \|\beta - \alpha\| \leq \|\beta - r\|$ if and only if $\beta - \alpha \in W^\perp$.
- 2). If such vector in 1) exists, then it is unique.
- 3). If W is finite-dimensional with an orthonormal basis $\{\alpha_1, \dots, \alpha_n\}$, then

$$\alpha = \sum_{i=1}^n \frac{\langle \beta, \alpha_i \rangle}{\|\alpha_i\|^2} \alpha_i.$$

is the unique best approximation.

Proof. Suppose that for all $r \in W, \|\beta - \alpha\| \leq \|\beta - r\|$. Then,

$$0 \leq \|\beta - r\|^2 - \|\beta - \alpha\|^2 = \|\alpha - r\|^2 + 2 \operatorname{Re} \langle \beta - \alpha, \alpha - r \rangle \quad \text{for all } r \in W.$$

Take for any $r \in W, r = \alpha - (\alpha - r)$, we can write

$$\|r\|^2 + 2 \operatorname{Re} \langle \beta - \alpha, r \rangle \geq 0 \quad \text{for all } r \in W$$

Now, for any $\mu \in W \setminus \{\alpha\}$, take $r = \frac{-\langle \beta - \alpha, \alpha - \mu \rangle}{\|\alpha - \mu\|^2} \cdot (\alpha - \mu)$. Then,

$$\frac{|\langle \beta - \alpha, \alpha - \mu \rangle|^2}{\|\alpha - \mu\|^2} - 2 \frac{|\langle \beta - \alpha, \alpha - \mu \rangle|^2}{\|\alpha - \mu\|^2} \geq 0$$

it holds iff $\langle \beta - \alpha, \alpha - \mu \rangle = 0$. Therefore $\langle \beta - \alpha, r \rangle = 0$ for all $r \in W$.

Conversely, if $\beta - \alpha \in W^\perp$, then for any $r \in W$,

$$\|\beta - r\|^2 = \|\beta - \alpha\|^2 + \|\alpha - r\|^2 + 2 \operatorname{Re} \langle \beta - \alpha, \alpha - r \rangle = \|\beta - \alpha\|^2 + \|\alpha - r\|^2 \geq \|\beta - \alpha\|^2.$$

And, the uniqueness is given by: if $\alpha_1, \alpha_2 \in W$ satisfies $\beta - \alpha_1, \beta - \alpha_2 \in W^\perp$, then

$$\text{for any } r \in W, \langle \beta - \alpha_1, r \rangle = \langle \beta - \alpha_2, r \rangle = 0$$

$$\Rightarrow \langle \alpha_1 - \alpha_2, r \rangle = 0$$

since $\alpha_1 - \alpha_2 \in W, \langle \alpha_1 - \alpha_2, \alpha_1 - \alpha_2 \rangle = 0$, thus $\alpha_1 = \alpha_2$.

And, the last assertion is proved by: for each $1 \leq k \leq n$,

$$\langle \beta - \alpha, \alpha_k \rangle = \langle \beta, \alpha_k \rangle - \langle \alpha, \alpha_k \rangle = 0, \text{ therefore } \beta - \alpha \in W^\perp.$$

□

Def. If for all $\beta \in V$, there is such $\alpha \in W$, then

$$E: V \rightarrow V: \beta \mapsto \alpha$$

is orthogonal projection.

Orthogonal Projection - Revised.

Let V be an inn. prod space, (not necessary finite dimension) and $W \subseteq V$ be a subspace. For any vector $\beta \in V$, if $\alpha \in W$ satisfies that:

$$\|\beta - \alpha\| \leq \|\beta - \gamma\| \quad \text{for all } \gamma \in W \quad \dots (*)$$

Then, the α is called best approximation to β by W , simply b.app.

Thm. $\alpha \in W$ is b.app to $\beta \in V$ by W iff for all $\mu \in W$, $\langle \beta - \alpha, \mu \rangle = 0$.

Proof. For fixed $\alpha \in W$ and $\beta \in V$, for any $\gamma \in W$,

$$\|\beta - \gamma\|^2 = \|\beta - \alpha + \alpha - \gamma\|^2 = \|\beta - \alpha\|^2 + \|\alpha - \gamma\|^2 + 2 \cdot \operatorname{Re} \langle \beta - \alpha, \alpha - \gamma \rangle$$

Therefore, the condition (*) holds if and only if

$$0 \leq \|\alpha - \gamma\|^2 + 2 \cdot \operatorname{Re} \langle \beta - \alpha, \alpha - \gamma \rangle \quad \text{for all } \gamma \in W.$$

Now take $\gamma = \alpha + \frac{\langle \beta - \alpha, \alpha - \mu \rangle}{\|\alpha - \mu\|^2} (\alpha - \mu)$ for any $\mu \in W$ s.t.

$$\text{Then, } 0 \leq \frac{|\langle \beta - \alpha, \alpha - \mu \rangle|^2}{\|\alpha - \mu\|^2} - 2 \cdot \frac{|\langle \beta - \alpha, \alpha - \mu \rangle|^2}{\|\alpha - \mu\|^2} = - \frac{|\langle \beta - \alpha, \alpha - \mu \rangle|^2}{\|\alpha - \mu\|^2}$$

And the inequality holds iff $\langle \beta - \alpha, \alpha - \mu \rangle = 0$, and all non zero vectors in W can be represented by $\alpha - \mu$, thus proved.

Conversely, if $\beta - \alpha$ is orthogonal to every vector in W , then $\operatorname{Re} \langle \beta - \alpha, \alpha - \gamma \rangle = 0$, thus $\|\beta - \gamma\|^2 - \|\beta - \alpha\|^2 = \|\alpha - \gamma\|^2 \geq 0$. $\square$

Thm. If b.app to $\beta \in V$ by W exists, then it is unique.

Proof. Suppose that $\alpha_1, \alpha_2 \in W$ satisfies for all $\gamma \in W$,

$$\langle \beta - \alpha_1, \gamma \rangle = \langle \beta - \alpha_2, \gamma \rangle = 0.$$

Then, take $\gamma = \alpha_1 - \alpha_2$, so $\langle \alpha_1 - \alpha_2, \alpha_1 - \alpha_2 \rangle = 0$. $\square$

Def. For all $\beta \in V$, there is b.app $\alpha \in W$, then the map

$$P: V \rightarrow V: \beta \mapsto \alpha$$

is called an orthogonal projection on W .

Thm. Let W be a fin. dim subspace of inn. prod space V .

Then, the orthogonal projection $P: V \rightarrow V$ exists. Furthermore,

P is idempotent linear operator with $\text{Im } P = W$, $\text{Ker } P = W^\perp$.

Proof. Let $\{\alpha_i, \dots, \alpha_n\}$ be an orthonormal basis for W . Then, a map

$$P: V \rightarrow V; \beta \mapsto \sum_{i=1}^n \frac{\langle \beta, \alpha_i \rangle}{\|\alpha_i\|^2} \cdot \alpha_i$$

1). It maps β to orthogonal projection by W .

For each $1 \leq i \leq n$,

$$\left\langle \beta - \sum_{i=1}^n \frac{\langle \beta, \alpha_i \rangle}{\|\alpha_i\|^2} \alpha_i, \alpha_i \right\rangle = \langle \beta, \alpha_i \rangle - \langle \beta, \alpha_i \rangle = 0$$

So, $\beta - P(\beta) \in W^\perp$ for all $\beta \in V$. (And, such a vector is unique, thus well-defined)

2). It is linear.

Let $\beta, \gamma \in V$ and $c \in F$ be given. Then,

$$P(c\beta + \gamma) = \sum \frac{\langle c\beta + \gamma, \alpha_i \rangle}{\|\alpha_i\|^2} \cdot \alpha_i = c \cdot \sum \frac{\langle \beta, \alpha_i \rangle}{\|\alpha_i\|^2} \alpha_i + \sum \frac{\langle \gamma, \alpha_i \rangle}{\|\alpha_i\|^2} \alpha_i = cP(\beta) + P(\gamma)$$

3). It is idempotent.

Given $\beta \in V$, $P(\beta) \in W$. So, $P^2(\beta) = P(\beta)$.

4). $\text{Im } P = W$, $\text{Ker } P = W^\perp$.

$\text{Im } P \subseteq W$ is clear, and for any $r \in W$, $P(r) = r \in \text{Im } P$. And,

$$\beta \in \text{Ker } P \Leftrightarrow P(\beta) = 0 \Leftrightarrow \langle \beta, \alpha_i \rangle = 0 \text{ for all } 1 \leq i \leq n \Leftrightarrow \beta \in W^\perp.$$

Note: Let $E: V \rightarrow V$ be a projection. In general, E can not be determined by its image.

For example, let $V = \mathbb{R}^3$, $W_1 = \text{Span}\{(1, 0, 0)\}$, $W_2 = \text{Span}\{(0, 1, 0), (0, 0, 1)\}$, $W_3 = \text{Span}\{(0, 1, 0), (1, 1, 1)\}$.

Then, $V = W_1 \oplus W_2 = W_1 \oplus W_3$, but $W_2 \neq W_3$. Moreover,

$$E: V \rightarrow V; (x_1, x_2, x_3) \mapsto (x_1, 0, 0), \quad E': V \rightarrow V; (x_1, x_2, x_3) \mapsto (x_1 - x_3, 0, 0)$$

Both E and E' are projections with $\text{Im } E = \text{Im } E' = \text{Span}\{(1, 0, 0)\}$, but $E \neq E'$,

$\text{Ker } E = W_2$ and $\text{Ker } E' = W_3$.

Corollary. Let $\{\alpha_i, \dots, \alpha_n\}$ be an orthogonal set of non-zero vectors in inn. prod space V . Then,

$$\text{for all } \beta \in V, \quad \sum_{i=1}^n \frac{|\langle \beta, \alpha_i \rangle|^2}{\|\alpha_i\|^2} \leq \|\beta\|^2, \text{ the equality holds iff } \beta = \sum \frac{\langle \beta, \alpha_i \rangle}{\|\alpha_i\|^2} \cdot \alpha_i.$$

Proof. Take $\gamma = \sum [\langle \beta, \alpha_i \rangle / \|\alpha_i\|] \cdot \alpha_i$. Then, $\langle \beta - \gamma, \gamma \rangle = 0$. So

$$\|\beta\|^2 = \|\gamma\|^2 + \|\beta - \gamma\|^2 \geq \|\gamma\|^2.$$

Riesz representation Theorem.

Thm. Let V be a fin. dim. inn. prod. v.s.

For any linear functional $f: V \rightarrow F$, there is a unique vector $\beta \in V$ s.t.

$$\text{for all } \alpha \in V, \quad f(\alpha) = \langle \alpha, \beta \rangle$$

In other words, $\mathcal{Q}: V \rightarrow V^*; \beta \mapsto \langle \cdot, \beta \rangle$ is a Conjugate isomorphism.

Proof. Let $\{\alpha_1, \dots, \alpha_n\}$ be an orthonormal basis for V .

Given linear functional $f: V \rightarrow F$, put

$$\beta = \sum_{j=1}^n \overline{f(\alpha_j)} \cdot \alpha_j.$$

then, for each $1 \leq k \leq n$,

$$\langle \alpha_k, \beta \rangle = \left\langle \alpha_k, \sum_{j=1}^n \overline{f(\alpha_j)} \alpha_j \right\rangle = \sum_{j=1}^n \overline{f(\alpha_j)} \cdot \langle \alpha_k, \alpha_j \rangle = \overline{f(\alpha_k)}$$

Clearly, $\langle \cdot, \beta \rangle$ is linear, thus the existence is proved. And the uniqueness is shown:

if $\langle \alpha, \beta \rangle = \langle \alpha, \gamma \rangle$ for all $\alpha \in V$, then for $\alpha = \beta - \gamma$, $\langle \beta - \gamma, \beta - \gamma \rangle = 0$.

Thus, $\beta = \gamma$. $\square$

Observe that a inn. prod. space V with orthonormal basis $\{\alpha_1, \dots, \alpha_n\}$

Let $\alpha = x_1 \alpha_1 + \dots + x_n \alpha_n$ and $\beta = y_1 \alpha_1 + \dots + y_n \alpha_n$. Then,

$$\langle \alpha, \beta \rangle = x_1 \overline{y_1} + \dots + x_n \overline{y_n}$$

Meanwhile, for any linear functional $f \in V^*$,

$$f(\alpha) = f(\alpha_1) x_1 + \dots + f(\alpha_n) x_n$$

If we want to β satisfies $\langle \alpha, \beta \rangle = f(\alpha)$, then take $y_i = \overline{f(\alpha_i)}$. Then is,

$$\beta = \overline{f(\alpha_1)} \alpha_1 + \dots + \overline{f(\alpha_n)} \alpha_n$$

Adjoint.

Def. Let T be a linear operator on inn. Prod. space V .

If there is a linear operator T^* s.t.

$$\langle T(\alpha), \beta \rangle = \langle \alpha, T^*(\beta) \rangle \text{ for all } \alpha, \beta \in V$$

Then such T^* is called adjoint of T .

Thm. If T is a linear operator on fin. dim inn. Prod space V ,
then T has a unique adjoint map T^* .

Proof. By the Riesz representation Thm, given $\beta \in V$, there is a unique $\beta' \in V$ s.t.
for all $\alpha \in V$, $\langle T(\alpha), \beta \rangle = \langle \alpha, \beta' \rangle$.

Define $T^*: V \rightarrow V: \beta \mapsto \beta'$. To show the linearity, let $\beta, \gamma \in V$ and $c \in \mathbb{F}$ be given.
Then, for any $\alpha \in V$,

$$\begin{aligned} \langle \alpha, T^*(c\beta + \gamma) \rangle &= \langle T(\alpha), c\beta + \gamma \rangle = \overline{c} \langle T(\alpha), \beta \rangle + \langle T(\alpha), \gamma \rangle = \overline{c} \langle \alpha, T^*(\beta) \rangle + \langle \alpha, T^*(\gamma) \rangle \\ &= \langle \alpha, c T^*(\beta) \rangle + \langle \alpha, T^*(\gamma) \rangle = \langle \alpha, c T^*(\beta) + T^*(\gamma) \rangle. \end{aligned}$$

$$\text{So } T^*(c\beta + \gamma) = c T^*(\beta) + T^*(\gamma).$$

Thm. Let V be a fin. dim. inn. Prod. space and let $\mathcal{B} = \{d_1, \dots, d_n\}$ be an orthonormal basis.
For any linear operator on V ,

$$[T]_{\mathcal{B}} = \left(\langle T(d_j), d_i \rangle \right)_{i,j}.$$

Proof. For each $1 \leq i \leq n$,

$$T(d_i) = \sum_{j=1}^n \langle T(d_i), d_j \rangle \cdot d_j$$

Thus proved.

Adjoint vs Transpose.

$T: V \rightarrow V$.

Adjoint: $T^*: V \rightarrow V: \beta \mapsto$

Transpose: $T^t: V^* \rightarrow V^*: \alpha \mapsto \alpha \circ T$

If V is fin. dim, then $V^* = \{ \langle \cdot, \beta \rangle \mid \beta \in V \}$, so
 $\alpha \circ T = \langle \cdot, \beta \rangle \circ T = \langle T(\cdot), \beta \rangle$

s.t. $\langle T(\alpha), \beta \rangle = \langle \alpha, T^*(\beta) \rangle$ for all $\alpha, \beta \in V$

Preserves Inner Product.

Def Let V and W be Inn. prod spaces over F .

We say that a linear map $T: V \rightarrow W$ preserves inner product if for all $\alpha, \beta \in V$,

$$\langle T(\alpha), T(\beta) \rangle = \langle \alpha, \beta \rangle.$$

Lemma Let V and W be Inn. prod spaces over F with $\dim V = \dim W < \infty$.

For linear map $T: V \rightarrow W$, TFAE:

1). T preserves inner product.

2). T is isomorphism preserving inner product.

3). T maps every orthonormal basis for V onto orthonormal basis for W .

4). T maps a orthonormal $\parallel$

proof. 1) $\Rightarrow$ 2) $\|T(\alpha)\|^2 = \|\alpha\|^2$, so $\ker T = \{0\}$ 2) $\Rightarrow$ 1) is trivial.

2) $\Rightarrow$ 3) is also trivial, if $\{\alpha_1, \dots, \alpha_n\}$ is orthonormal basis for V , then $\{T(\alpha_1), \dots, T(\alpha_n)\}$ is too, $\langle T(\alpha_i), T(\alpha_j) \rangle = \langle \alpha_i, \alpha_j \rangle = \delta_{ij}$.

3) $\Rightarrow$ 4) $\Rightarrow$, 4) $\Rightarrow$ 1) Let $\{\alpha_1, \dots, \alpha_n\}$ be an orthonormal basis with

$\{T(\alpha_1), \dots, T(\alpha_n)\}$ also orthonormal basis for W . Then, for any $\alpha, \beta \in V$,

$\alpha = \sum x_i \alpha_i$, $\beta = \sum y_i \alpha_i$ f.s $x_i, y_i \in F$. Now,

$$\langle T(\alpha), T(\beta) \rangle = \langle \sum x_i \cdot T(\alpha_i), \sum y_j \cdot T(\alpha_j) \rangle = \sum_i \sum_j x_i \cdot \overline{y_j} \langle T(\alpha_i), T(\alpha_j) \rangle = \sum_i \sum_j x_i \cdot \overline{y_j} \delta_{ij} = \sum_i x_i \overline{y_i} = \langle \alpha, \beta \rangle.$$

Thm Let V and W be Inn. prod space over F , and let $T: V \rightarrow W$ be a linear map. Then,

T preserves inner product if and only if T preserves Norm.

proof. One direction is trivial. Suppose that T preserves Norm. By the polarization identity,

$$\begin{aligned} \langle T(\alpha), T(\beta) \rangle &= \frac{1}{4} \cdot \sum_{n=1}^4 i^n \cdot \|T(\alpha) + i^n \cdot T(\beta)\|^2 = \frac{1}{4} \cdot \sum_{n=1}^4 i^n \cdot \|T(\alpha + i^n \beta)\|^2 = \frac{1}{4} \cdot \sum_{n=1}^4 i^n \cdot \|\alpha + i^n \beta\|^2 \\ &= \langle \alpha, \beta \rangle. \quad \square \end{aligned}$$

Unitary Operator.

Def. Let V be an Inn. prod space.

An automorphism preserving inn. prod is called unitary operator.

Equivalently, U is unitary iff adjoint U^* exists and $U^* \circ U = U \circ U^* = I$.

Proof of the equivalence: Suppose that U is unitary. Then, for any $\alpha, \beta \in V$,

$$\langle U(\alpha), \beta \rangle = \langle U(\alpha), (U \circ U^{-1})(\beta) \rangle = \langle \alpha, U^{-1}(\beta) \rangle.$$

So $U^{-1} = U^*$. Conversely, suppose U^* exists and $U^* = U^{-1}$. Then invertible is clear, and

$$\langle U(\alpha), U(\beta) \rangle = \langle \alpha, (U^* \circ U)(\beta) \rangle = \langle \alpha, \beta \rangle. \quad \square$$

Def. A real or complex Matrix $A \in \text{Mat}_n$ is called orthogonal if $A^t A = I$.

Thm. For any $B \in GL_n(\mathbb{C})$, there exists a unique lower-triangular M with positive diagonal s.t. MB is unitary.

Proof. Since B is invertible, the row vectors $\{\beta_1, \dots, \beta_n\}$ is basis for $\mathbb{C}^n$.

Let $\{\alpha_1, \dots, \alpha_n\}$ be an orthonormal basis for $\mathbb{C}^n$ obtained from β_i 's by Gram-Schmidt.

Now, for each $1 \leq k \leq n$, $\alpha_k = \beta_k - \sum_{i < k} \frac{\langle \beta_k, \alpha_i \rangle}{\|\alpha_i\|^2} \cdot \alpha_i$. And since $\text{span}\{\beta_1, \dots, \beta_{k-1}\} = \text{span}\{\alpha_1, \dots, \alpha_{k-1}\}$,

$$\alpha_k = \beta_k - \sum_{i < k} \overset{\text{unique}}{C_{ki}} \beta_i \quad \text{for some } C_{ki} \in \mathbb{C}.$$

$$\text{Let } U = \begin{bmatrix} \frac{\alpha_1}{\|\alpha_1\|} & \dots & \frac{\alpha_n}{\|\alpha_n\|} \end{bmatrix} \quad \text{and} \quad M_{ki} = \begin{cases} -\frac{1}{\|\alpha_k\|} \cdot C_{ki} & \text{if } i < k \\ \frac{1}{\|\alpha_k\|} & \text{if } i = k \\ 0 & \text{otherwise.} \end{cases}$$

↑ Revised in After page.

Unique decomposition Theorem. (OR QR decomposition)

For any $B \in GL_n(\mathbb{C})$, there exists a unique lower-triangular M with positive diagonal s.t MB is unitary,

Proof. Since B is invertible, the row vectors of B , $\{\beta_1, \dots, \beta_n\}$ is basis for V .

Apply Gram-Schmidt process, that is, for each $1 \leq k \leq n$, put

$$\alpha_k = \beta_k - \sum_{j < k} \frac{\langle \beta_k, \alpha_j \rangle}{\|\alpha_j\|^2} \alpha_j$$

Then, $\text{span}\{\alpha_1, \dots, \alpha_r\} = \text{span}\{\beta_1, \dots, \beta_r\}$ by induction, and $\{\alpha_1, \dots, \alpha_n\}$ is orthonormal.

Put $U = \begin{bmatrix} \frac{\alpha_1}{\|\alpha_1\|} & \dots & \frac{\alpha_n}{\|\alpha_n\|} \end{bmatrix}$, then clearly $U^* U = I$, because $\{\alpha_1, \dots, \alpha_n\}$ is orthonormal

And, for each $1 \leq k \leq n$, $\alpha_k = \beta_k - \sum_{j < k} C_{kj} \beta_j$ for some C_{kj} .

Put $M_{kj} = \begin{cases} -\frac{C_{kj}}{\|\alpha_k\|} & \text{if } k > j \\ \frac{1}{\|\alpha_k\|} & \text{if } k = j \\ 0 & \text{otherwise.} \end{cases}$ Then, each $1 \leq k \leq n$, $U_k = \begin{pmatrix} & & 0 \\ & & \\ M_{k1} & \dots & M_{kn} \end{pmatrix} \begin{pmatrix} \beta_1 & \dots & \beta_n \end{pmatrix} = U$

So $MB = U$. To show the uniqueness, denote

$$T^+(n) = \{M \in M_{nn}(\mathbb{C}) \mid M \text{ is lower triangular, positive diagonals}\}.$$

Let $M_1, M_2 \in T^+(n)$ satisfies $M_1 B, M_2 B$ are unitary. Since unitary matrices forms a group

$$(M_1 B) \cdot (M_2 B)^{-1} = M_1 M_2^{-1} \text{ is also unitary.}$$

And, M_2^{-1} is lower-triangular with positive diagonals, so is $M_1 M_2^{-1}$. Now,

$$(M_1 M_2^{-1})^{-1} = (M_1 M_2^{-1})^*, \text{ so } M_1 M_2^{-1} \text{ is diagonal.}$$

So, $M_1 M_2^{-1} = I$, or $M_1 = M_2$. @

Corollary. For each $B \in GL(n)$, there exists unique matrices K, A, N s.t $B = KAN$,

where K is unitary, A is diagonal with positive values, N is upper-triangular with diagonals 1

+ Note. Let $T_n(F)$ be the set of $n \times n$ upper-triangular matrices with all diagonals are 1.

Then, $(T_n(F), \cdot)$ is group, because: for any $A \in T_n(F)$, $A = I + N$,

where N is strictly upper triangular with $N^n = 0$, nilpotent. So,

$$A^{-1} = I - N + N^2 - N^3 + \dots + (-N)^{n-1},$$

and $A \in T_n(F)$ is clear.

QR - Factorization.

Let $A = (w_1 \dots w_n)$ be an invertible matrix.

And, for each $1 \leq k \leq n$,
$$V_k = w_k - \sum_{i=1}^{k-1} \frac{\langle w_k, V_i \rangle}{\|V_i\|^2} \cdot V_i.$$

Then, $\{V_1, \dots, V_n\}$ is an orthogonal, so put $U_k = V_k / \|V_k\|$, then

$\{U_1, \dots, U_n\}$ is an orthonormal. Clearly, $Q \stackrel{\text{def}}{=} (U_1 \dots U_n)$ is unitary.

Moreover, for each $1 \leq k \leq n$,

$$w_k = \|V_k\| \cdot U_k + \sum_{i=1}^{k-1} \langle w_k, U_i \rangle \cdot U_i.$$

Define

$$R_{jk} = \begin{cases} \|V_k\| & j=k \\ \langle w_k, U_j \rangle & j < k \\ 0 & j > k \end{cases}$$

Then, R is upper-triangular, and satisfying

$$Q^{-1}A = Q^*A = \begin{pmatrix} \overline{u_1} \\ \vdots \\ \overline{u_n} \end{pmatrix} \cdot (w_1 \dots w_n) = \begin{pmatrix} \langle w_1, u_1 \rangle & \langle w_2, u_1 \rangle & \dots & \langle w_n, u_1 \rangle \\ \langle w_1, u_2 \rangle & - & - & \vdots \\ \vdots & - & - & - \\ \langle w_1, u_n \rangle & - & - & \langle w_n, u_n \rangle \end{pmatrix} \\ = R.$$

so, $A = QR$.

Matrix Decomposition.

In this paper, our ambient space is $M_{n \times n}(F)$, simply M .

Given $A \in M$, denote $A^{[i]}$ is i 'th row of A , $A_{[j]}$ is j 'th column of A .

With standard inner product on F^n , we can write

$$A \cdot B = \left(\langle A^{[i]}, B^{[j]} \rangle \right)_{i,j}.$$

$$\text{and } \langle A^{[i]}, B^{[j]} \rangle = \sum_{k=1}^n A_{ki} \cdot B_{kj}$$

T -Cyclic subspace.

In this, every vector space is finite-dimensional, and $T: V \rightarrow V$ is arbitrary linear operator.

Def. Given a vector $\alpha \in V$, define T -cyclic subspace generated by α is:

$$Z(\alpha; T) \stackrel{\text{def}}{=} \{g(T)(\alpha) \mid g(t) \in F[t]\} = \text{span}_F \{T^k(\alpha) \mid k \geq 0\}$$

If $Z(\alpha; T) = V$, then α is called a cyclic vector for T .

Def. Given $\alpha \in V$, the kernel of a map

$$\varphi_T^\alpha: F[t] \rightarrow V; g(t) \mapsto g(T)(\alpha)$$

is called T -annihilator of α , and since $F[t]$ is P.I.D., it has unique monic generator.

Denote $\ker \varphi_T^\alpha = M(\alpha; T) = (P_\alpha)$ where P_α is monic.

And, P_α divides the minimal polynomial of T .

Thm. Given non-zero vector $\alpha \in V$, and T -annihilator of α , the followings hold:

- 1). $\dim Z(\alpha; T) = \deg P_\alpha$.
- 2). $Z(\alpha; T) = \text{span}_F \{T^i(\alpha) \mid 0 \leq i \leq k-1\}$ where $k = \deg P_\alpha$.
- 3). The restriction $T|_{Z(\alpha; T)}$ has a minimal polynomial as P_α .

Proof. Let $g(t) \in F[t]$ be given. Then, by the division algorithm,

$$g = P_\alpha \cdot q + r \quad (r=0 \text{ or } \deg r < \deg P_\alpha)$$

So, $g(T)(\alpha) = (P_\alpha \cdot q)(T)(\alpha) + r(T)(\alpha) = r(T)(\alpha) \in \text{span}_F \{T^i(\alpha) \mid 0 \leq i \leq k-1\}$.

And, these are linearly independent, by the minimality of P_α .

To show the last assertion, check that $Z(\alpha; T)$ is T -invariant. Now, given $g \in F[t]$,

$$(P_\alpha(T|_Z) \circ g(T))(\alpha) = (P_\alpha(T) \circ g(T))(\alpha) = g(T)(\alpha) = 0.$$

And, by the minimality of P_α , it is also minimal polynomial of $T|_Z$.

The Companion Matrix.

Let U be a linear operator on W of dimension $k \in \mathbb{N}$, which has a cyclic vector α .

Let $\alpha_i = U^{i-1}(\alpha)$, $i=1, \dots, k$. Then, $\{\alpha_1, \dots, \alpha_k\}$ is ordered basis for U , and

$\alpha_{i+1} = U^i(\alpha) = U(\alpha_i)$, $i=1, \dots, k-1$, and for minimal polynomial p_α of U ,

$$p_\alpha(U)(\alpha) = U^k(\alpha) + c_{k-1}U^{k-1}(\alpha) + \dots + c_1U(\alpha) + c_0\alpha = 0.$$

So $U^k(\alpha) = U(\alpha_k) = -c_0\alpha_1 - c_1\alpha_2 - \dots - c_{k-1}\alpha_k$. Now, for $\mathcal{B}$

$$[U]_{\mathcal{B}} = \begin{bmatrix} 0 & 0 & \dots & 0 & -c_0 \\ 1 & 0 & & 0 & -c_1 \\ 0 & 1 & & 0 & \vdots \\ \vdots & 0 & \ddots & \vdots & \vdots \\ 0 & 0 & \dots & 1 & -c_{k-1} \end{bmatrix} \text{ is called the Companion Matrix of } p_\alpha.$$

Def. Let V be an f.n.d.m. inn. prod. space.

A linear operator $T: V \rightarrow V$ is called Normal if $T \circ T^* = T^* \circ T$.

+ Notes the existence of adjoint is guaranteed by the Riesz Representation Theorem.

Thm. Let V be an f.n.d.m. inn. prod. space, and T be a Hermitian.

Then, every eigenvalue of T is real, and eigenvectors of distinct eigenvalues are orthogonal.

Proof. Given eigenvalue c of T , for any eigenvector $\alpha \in V$ corresponding to c , observe

$$c \cdot \langle \alpha, \alpha \rangle = \langle c\alpha, \alpha \rangle = \langle T(\alpha), \alpha \rangle = \langle \alpha, T^*(\alpha) \rangle = \langle \alpha, T(\alpha) \rangle = \langle \alpha, c\alpha \rangle = \overline{c} \cdot \langle \alpha, \alpha \rangle$$

$\xrightarrow{\text{Def}} \quad \xrightarrow{T\alpha=c\alpha} \quad \xrightarrow{V.f.n. \text{ adj. exist}} \quad \xrightarrow{\text{Hermitian}} \quad \xrightarrow{T\alpha=c\alpha} \quad \xrightarrow{\text{Def}}$

Since $\langle \alpha, \alpha \rangle$ is positive real number ($\because \alpha \neq 0$), so $c = \overline{c}$, or $c \in \mathbb{R}$.

Moreover, let $c_1, c_2 \in \mathbb{R}$ be distinct eigenvalues, and α_1, α_2 be eigenvectors of c_1, c_2 , respectively. Then,

$$c_1 \cdot \langle \alpha_1, \alpha_2 \rangle = \langle c_1 \alpha_1, \alpha_2 \rangle = \langle T(\alpha_1), \alpha_2 \rangle = \langle \alpha_1, T^*(\alpha_2) \rangle = \langle \alpha_1, T(\alpha_2) \rangle = \langle \alpha_1, c_2 \alpha_2 \rangle = \overline{c_2} \langle \alpha_1, \alpha_2 \rangle$$

By the first assertion, $\overline{c_2} = c_2$, so $(c_1 - c_2) \cdot \langle \alpha_1, \alpha_2 \rangle = 0$.

Finally, $\mathbb{C}$ has no zero divisor, $\langle \alpha_1, \alpha_2 \rangle = 0$. (A kind of take.)

Thm. Every Hermitian on f.n.d.m. inn. prod. space has an eigenvalue.

Proof. Set V be a such space, and T is Hermitian. Take $\mathcal{B}$ be an orthonormal basis for V .

Let $A = [T]_{\mathcal{B}}$. Then, $A = [T^*]_{\mathcal{B}} = [T]_{\mathcal{B}}^* = A^*$.

Put $W \stackrel{\text{def}}{=} \{X \in M_{n \times 1}(\mathbb{C})\}$, with inner product $\langle X, Y \rangle = Y^* X$.

Then, a map $T: W \rightarrow W: X \mapsto AX$ is Hermitian. Moreover, the characteristic polynomial of T .

$$\det(xI - T) = \det(xI - A)$$

is a polynomial of degree n over $\mathbb{C}$, so it has a root $c \in \mathbb{C}$.

Thus, c is an eigenvalue of T , Hermitian, so it is a real number.

Note: Characteristic Polynomial of Hermitian splits over the given field $\mathbb{R}$ or $\mathbb{C}$.

Lemma. Let V be an fin. dim inn prod space, and T be a linear operator.

For any T -invariant subspace $W \subset V$, the orthogonal complement $W^\perp$ is T^* -invariant.

Proof. $W^\perp$ is T^* -invariant iff for any $\alpha \in W^\perp$, $T(\alpha) \in W^\perp$.

Given $\alpha \in W^\perp$, $\langle \alpha, T(\beta) \rangle = 0$ for all $\beta \in W$, since $T(\beta) \in W$.

That is, $\langle T^*(\alpha), \beta \rangle = 0$ for all $\beta \in W$, or $T^*(\alpha) \in W^\perp$.

Thm. Let V be an fin. dim inn prod space, and let $T: V \rightarrow V$ be a linear operator.

T is Hermitian if and only if There is an orthonormal basis for V consisted by eigenvectors of T .

Proof. Using induction for $\dim V = n$.

Since T is Hermitian, it has at least one eigenvector $\alpha_1 \in V$. Now,

$\{\alpha_1 / \|\alpha_1\|\}$ is the orthonormal basis we desired. Now, let the statement hold for $n-1$.

Let $\dim V = n$. Then